Conditional objective bounds from physical threshold counts

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Round141 independent research appendix, 30 October 2026. This section extends the maintained Round137 inventory-Gini routing manuscript and the Round138-Round140 research increments. The complete model, physical objective and stable GI-TBC method are retained. The study asks whether a station-count axis supplies information beyond the original Gini axis, and whether qualified conditional information improves a complete method. It is a finite study of known research configurations, not an independent validation set.

1. Integer count domains and conditional numerator chords

Only stations are counted. For actual finite inventory support $\mathcal{Y}_i$, let $\sum_y s_{iy} = 1$, $Y_i = \sum_y y s_{iy}$, and $r_i = c_i Y_i$. The positive distinct supported ratios form $0 = t_0 < t_1 < \cdots < t_M$, with $\Delta_j = t_j - t_{j-1}$. Define

$$K_j = \sum_i \sum_{y: c_i y \geq t_j} s_{iy}, \qquad A = \sum_{i<l} |r_i - r_l|.$$

For integer selectors, each threshold separates exactly $K_j(n-K_j)$ station pairs. Summing their threshold lengths proves

$$A = \sum_j \Delta_j K_j (n - K_j).$$

Round140 obtains valid integer ranges $\ell_j \le K_j \le u_j$ from actual supports and the physical inventory budgets $\max\{0, B - \sum_k Q_k\} \le \sum_i Y_i \le B$. For the inherited numerator epigraph H_Σ, TC2 retains the old TC row and adds the concave chord

$$H_\Sigma \ge \sum_j \Delta_j[(n - \ell_j - u_j)K_j + \ell_j u_j].$$

The remainder of one term is $(K_j - \ell_j)(u_j - K_j)$. It measures chord slack, not complete-model objective error. Choose $\ell_j \le m < u_j$ and partition a physical parent domain into $D_0 = D \cap \{K_j \le m\}$ and $D_1 = D \cap \{K_j \ge m + 1\}$. Integer counts make this a mutually exclusive complete cover. The sides are not two Gini intervals.

Nested threshold counts imply the zero-Optimize propagation

$$D_0 :\ u'_h \leftarrow \min(u_h, m)\ (h \ge j), \qquad D_1 :\ \ell'_h \leftarrow \max(\ell_h, m + 1)\ (h \le j).$$

Each restricted model appends the resulting conditional TC2 aggregate. Parent TC and TC2 remain. Integer-valid ranges need not already constrain every fractional parent point, so replacement would not justify unconditional LP dominance. Incompatible intervals are not clipped into a valid range. If $d = u_j - \ell_j$ and $m = \ell_j + \lfloor d/2 \rfloor$, each child width is at most $\lfloor d/2 \rfloor$. The selected term's continuous worst remainder decreases from $\Delta_j d^2/4$ to at most $\Delta_j d^2/16$. This is not a fourfold BRP bound or speed theorem.

2. Qualified side floors and a scalar disjunction

Suppose a qualified relaxation of side D_i, intersected with the improvement cutoff $F \le c_i$, gives lower bound λ_i. Use $\beta_i \le \min\{\lambda_i, c_i\}$, with the inherited fixed numerical margin and downward arithmetic. A physical solution below the cutoff has a legal extension to this model; one above the cutoff already has $F \ge c_i$. This proves the floor on the entire corresponding physical side within its source support/domain. Qualified infeasibility below the cutoff may supply its conservative cutoff floor. It does not prove that the whole physical side is empty, delete that side, or fix a binary. Unqualified returns do not supply new floors. An incomplete pair keeps ordinary valid parent information unless a source rejection requires the existing fail-closed disposition. The complete count-cover floor is $\min\{\beta_0, \beta_1\}$.

One binary b links the physical count to the two sides:

$$K_j \le m + (u_j - m)b, \qquad K_j \ge \ell_j + (m + 1 - \ell_j)b.$$

The complete linear objective, including ObjCon, receives

$$\text{D1}: \quad F \ge \beta_0 + (\beta_1 - \beta_0)b.$$

D0 independently pays the same selection and proofs and keeps both links and the binary, replacing only this row by $F \ge \min\{\beta_0, \beta_1\}$. Each physical integer improvement solution in the source support/domain chooses its unique side from inventory Y, not from a virtual auxiliary Gini value. Both signs of the floor difference and nonzero ObjCon receive directed endpoint guards. The serialized endpoints do not exceed the qualified floors; outward rounding can weaken the ideal scalar formula.

For fixed $K \in [\ell, u]$, feasible fractional binary values satisfy

$$\max\{0, (K - m)/(u - m)\} \le b \le \min\{1, (K - \ell)/(m + 1 - \ell)\}.$$

When $\beta_1 \geq \beta_0$, eliminating b gives $F \geq \beta_0$ and $F \geq \beta_0 + (\beta_1 - \beta_0)(K - m)/(u - m)$. In the reverse case it gives $F \geq \beta_1$ and $F \geq \beta_1 + (\beta_0 - \beta_1)(m + 1 - K)/(m + 1 - \ell)$. Together with the count interval, these describe the convex hull of the two scalar interval epigraphs $[\ell, m] \times [\beta_0, \infty)$ and $[m + 1, u] \times [\beta_1, \infty)$. This is a standard disjunctive device (Balas 1998), not the full BRP hull.

For $(\ell, u, m, \beta_0, \beta_1) = (0, 4, 1, 0, 2)$, the scalar point $(K, F, b) = (2, 0, 1/3)$ satisfies D0 and violates D1. The point $(2, 1, 1/2)$ satisfies continuous D1 but admits no binary D1 extension. These are scalar witnesses only. A chosen-b violation is not a projection separation test; all feasible fractional binary values must be considered.

3. Matched continuous evidence

Five saved-point scores reproduce the supplied goal without new Optimize. Three domains enter 39 new continuous solves: three parent controls, 24 plain/conditional count children, six common-source D0/D1 roots and six fully rebuilt Gini-midpoint children. All use explicit Seed0, one thread and serial timing. Ordinary MPS is serialized, hashed and read back before Optimize. These are numerical LP optima, not rational certificates or Native MIP root bounds. Offline diagnostic sources never enter a complete method.

K35's physical threshold is $\frac{9700060735874915}{9007199254740992}$; the other two use $\frac{18014398509481983}{18014398509481984}$. The latter is close to one but cannot be replaced by decimal one. Count/D0/D1 comparisons fix the source objective, support, Gini interval and actual serialized cutoff. The Gini comparison instead verifies an identical full parent, then rebuilds its actual restricted children, including their support and product rows.

The threshold maximizes the weighted chord remainder; the smaller physical threshold breaks a tie. Explicit count-range violations precede that score. Balanced and LP-floor pivots are compared. LP-floor has the stronger conditional cover in all three domains and is frozen as the uniform online rule. Online scoring uses long double; threshold membership and dyadic coefficient arithmetic are exact. It is not an exact-rational online score ordering claim. The following first table uses raw optimal values.

Role	Parent L	Plain LP-floor cover	Conditional balanced	Conditional LP-floor
K35	0.226044051	0.226044226	0.235747225	0.240140105
RB35-T2	0.135449235	0.135463277	0.142324212	0.146301753
K20	0.234821878	0.23483382	0.251586407	0.255556703

A balanced plain side retains the saved parent optimum. K35's plain differences at roughly 1e-7 are numerical scale, not substantive gains. The LP-floor integer gap can give small plain gains on RB/K20, while conditional chords produce the larger increments. All qualified covers below use the same conservative floor policy.

Role	Count conditional cover	Gini midpoint cover	Difference
K35	0.240140005	0.227601298	0.0125387073
RB35-T2	0.146301653	0.135641398	0.010660255
K20	0.255556603	0.235396129	0.0201604738

This establishes finite new count-axis information beyond the matched midpoint, not dominance over every Gini pivot or domain. D0 and D1 next use exactly the same two qualified side floors.

Role	beta0	beta1	D0 optimum	D1 optimum	Projection violation
K35	0.241539846	0.240140005	0.240140005	0.240140005	0
RB35-T2	0.146301653	0.150566842	0.146301653	0.146301653	0.000804724405
K20	0.255556603	0.268192485	0.255556603	0.255556603	0.00266927223

D1 separates two actual D0 optima after minimizing over fractional b, but admits other points with the same objective; no reoptimized common-root objective gain is observed. Both floors are already below the carrier's observed Native early bounds (approximately .44698 for K20, .15862 for RB35-T2, and .26602 for K35). That limits expected direct bound value, without proving global redundancy across different Native paths. The cover/projection evidence and small probe cost justify the finite pilot.

4. Complete methods and attribution

The uniform carrier is BOTH stock + old TC + TC2. Every new method pays complete own H (seed20260626), normal root/lookahead, construction, two count proofs, validation and Native search. Gurobi Seed0 is explicit for primary solves; any limited Seed1 keeps its registered identity. Threads1, affinity4, M-B/VD-P, pair-only, AM=.08, depth8, minimum width1e-4 and AP off remain common. K roles retain lambda=.15; resource roles retain rho=1 and the frozen binary64 lambda=1/Omega. Physical T is separate from experimental cap.

Role	Stations / vehicles	Physical T s	lambda	Cap s
K20	20 / 2	5400	0.15	1200
K35	35 / 3	10800	0.15	2400
K15	15 / 2	3600	0.15	600
RB35-T2	35 / 3	7200	0.0551252752	2400

The four D1 Seed0 roles retain two qualified certificates, K15 and K20, versus one for stable GI-TBC on these roles. STOCK, old TC+STOCK and TC2+STOCK already retain both proofs. K15 D1 produces its certificate in 32.3593 seconds (37.0207 total), compared with stock's 33.8182 (34.9299 total). K20 D1 closes in 93.5356 seconds (102.5815 total), compared with stock's 101.6354 (104.4597 total). Both differences against stock are below the inherited substantive threshold. K35 D1 ends at U/L=.655549726/.559530134: a substantial repair over TC2, but a stronger incumbent and weaker bound than stable GI-TBC; old TC+STOCK remains better at both ends. RB35-T2 D1 ends at .413773275/.227918419, worse at both ends than stock, TC2 and old TC+STOCK. Its U is about 20.1% worse than stable GI-TBC despite a stronger L. All formal positions are independently paid complete methods under the registered caps; open T_cert remains null.

Role	Method	Seed	Verified U	Supported L	T_cert s
K15	GI-TBC	0	0.592264882	0.592264879	112.769303
K15	STOCK	0	0.592264882	0.592264882	33.8182265
K15	TC+STOCK	0	0.592264882	0.59226488	45.2425381
K15	TC2+STOCK	0	0.592264882	0.592264882	36.1332107
K15	D1	0	0.592264882	0.592264882	32.3593347
K20	GI-TBC	0	0.716806546	0.705355793	open
K20	STOCK	0	0.716806546	0.716806546	101.635385
K20	TC+STOCK	0	0.716806546	0.716806546	135.696813
K20	TC2+STOCK	0	0.716806546	0.716806546	115.019623
K20	D0	0	0.716806546	0.716806546	117.225005
K20	D1	0	0.716806546	0.716806546	93.5355741
K35	GI-TBC	0	0.692854359	0.565765681	open
K35	STOCK	0	0.669595569	0.556776898	open
K35	TC+STOCK	0	0.650438712	0.568288526	open
K35	TC2+STOCK	0	0.702148613	0.541970377	open
K35	D0	0	0.667178614	0.5598416	open
K35	D1	0	0.655549726	0.559530134	open
RB35-T2	GI-TBC	0	0.344642626	0.21898595	open
RB35-T2	STOCK	0	0.401200218	0.2291816	open
RB35-T2	TC+STOCK	0	0.319080493	0.233927262	open
RB35-T2	TC2+STOCK	0	0.324275718	0.229904209	open
RB35-T2	D1	0	0.413773275	0.227918419	open

Open certificate times remain null; no cap is imputed as convergence. Historical references preserve actual original source/PE and complete clocks, pinned to accepted R140 science and its dependencies. They are not same-PE causal interventions. The actual panel and missing TC2 cells are not enlarged by inference. Both-certificate timing uses the inherited substantive threshold max(10 s, 10% of the smaller clock).

The same-PE K20 comparison isolates a useful conditional-floor effect: D1 saves 23.6894 production seconds and 24.7674 total seconds versus independently paid D0, with the same U and effectively identical certificate L. Its terminal Work/nodes are 137.2465/18588 versus D0's 173.2820/28554. K35 D1 improves D0's U by .011628889 but loses .000311467 in L, a real small bound regression; neither dominates. Their cap-limited producer clocks are effectively equal. D0 itself already improves TC2's K35 U by .0349700 and L by .0178712, so the whole carrier repair cannot be credited to the conditional delta. Exact root/probe identities, floors and first integer matrices except the verified floor row match in both pairs. No RB D0 was entered: its adverse whole-method result is not attributed causally to the conditional delta.

Role	D1-D0 U	D1-D0 L	D1-D0 produce s	D1-D0 total s
K20	0	1.66533454e-15	-23.6894309	-24.7673724
K35	-0.0116288886	-0.000311466832	-0.00239847181	0.036445328

For exposed pairs, root/probe matrices and floors match exactly. The actual first integer models are also identical after removing only the verified objective-floor row. This distinguishes D1's conditional value from D0's count links/common floor, while D0 versus TC2 measures their combined change.

All six formal methods actually select once, obtain both qualified count LPs and install the valid pair. There is no actual outer Gini split. D1 probe totals are about .323 seconds for K15, .594 for K20, 3.128 for K35 and 4.775 for RB35-T2, including construction/solve/qualification. The complete producer clocks pay these costs. The standalone D1/D0 common-root objectives coincide, even though the written fractional-b projection separates actual D0 optima on K20 and RB. Both side floors lie below the carrier's observed Native early bounds. The evidence therefore supports changed feasible geometry and search paths, without an additional root LP optimum, a general redundancy theorem, a claim that Native branches on b, or an additive allocation of runtime to a row.

K35 D1 uses slightly less terminal Work but more nodes than D0; it is not a universal search-effort improvement. The small probe cost does not by itself explain the important RB endpoint regression.

Role	Mode/Seed	Produce s	Verify s	Total s	Relative gap
K15	D1/0	32.3593347	4.6613453	37.02068	3.74907599e-15
K20	D0/0	117.225005	10.1238641	127.348869	2.47815376e-15
K20	D1/0	93.5355741	9.0459226	102.581497	1.5488461e-16
K35	D0/0	2371.13384	14.8301903	2385.96403	0.160881976
K35	D1/0	2371.13144	14.8690341	2386.00048	0.146471867
RB35-T2	D1/0	2371.05613	13.5478036	2384.60394	0.449170758

These are every new formal position; cap90 qualifications are separate. Full-precision U/L, signed gaps, Native returned ObjVal/ObjBoundC versus external merged L, root/probe/target/terminal times, Work, nodes and controls remain in the CSV/JSON authorities. Later independent reviews and supplemental chain checks are separate costs. Rounded logs do not establish exact physical-incumbent discovery clocks.

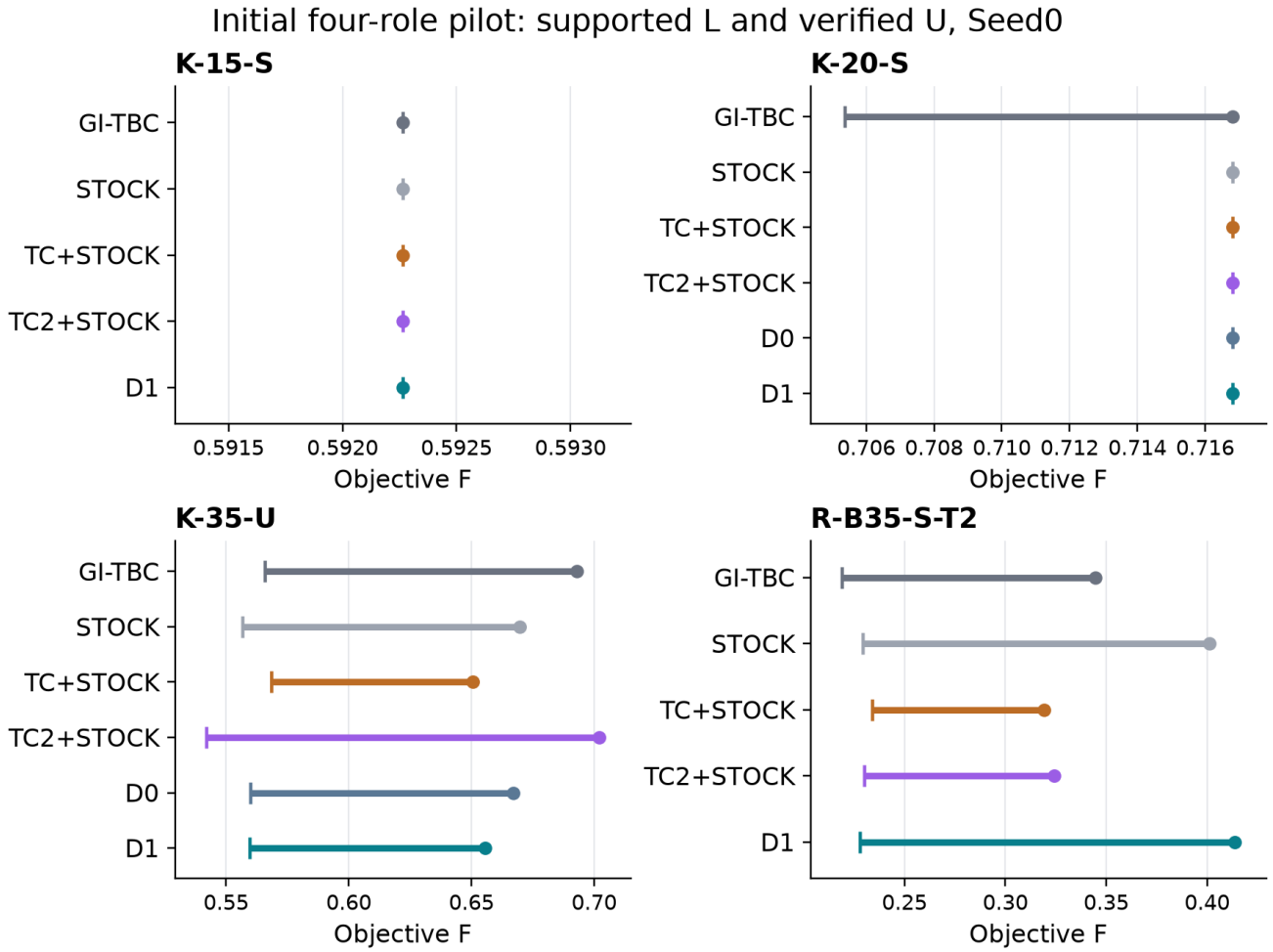


Figure 1: Initial four-role Seed0 pilot. Bars span supported L to verified U; they are not confidence intervals. Historical references retain their original identities. The complete tables include every entered position.

5. Implementation scope, history and decision

One root selection uses formal selector fields from the already required LP, not an observational CSV or extra duplicate Optimize. The two canonical count requests are independent local proofs, not scheduler children. Count settings survive parameter readback; independently verified final inventory determines witness side. A single side never becomes global/Gini-cover evidence. Source rejection invalidates dependent use under the inherited fail-closed policy. Later artifacts may inherit the pair only with the same input, contained Gini interval, nonexpanded cutoff and supported-inventory subsets, remapping the frozen physical threshold. Start b is computed from physical inventory. Original probe types are restored then each probe is immediately discarded. Default BranchPriority remains zero, without a custom branching callback or unobserved branching-on-b claim.

Finite tests cover both signs, nonzero ObjCon, both physical Start sides, same/opposite-side witnesses, source revocation, invalid scope, default-off reset, one contained successor and actual Native row/type reads. The first cap90 v5 producer failed the required disposable-probe type-restoration lifecycle check; raw/cost are retained outside performance. V6 fixes that lifecycle, followed by two healthy qualifications. Pure-reader corrections never rerun a sealed healthy producer. The controlled hybrid production build and final targeted repair retain their actual source/PE lineage.

R50/R51 studied variable-family priorities and sparse probing; R62 used single-station thresholds, R88 two-site CDF envelopes, and R120-R122 already used one-binary Gini-axis objective floors with mixed outcomes. The distinction here is an all-station physical count axis, integer gap, nested propagation and conditional numerator chord. McCormick/RLT and disjunctive constructions remain standard tools (McCormick 1976; Adams and Sherali 1990; Balas 1998).

The supplied same-objective R138 K35 numerical return-bound witness supersedes R140's prior priority suggestion. It has R'=(6.306613194,5.773547225,18.919839581), satisfies both stated return upper bounds with about 1.82e-12 maximum residual, and retains its full packet identity in the mathematical note. This is neither a general heterogeneous-vehicle symmetry theorem nor a rational optimality certificate, and is not a new Round141 Optimize or repeated replay. No return-product full benchmark is opened.

Independent result-based selection requires no further experiments. The remaining four D1 configurations are not entered after the important RB regression and lack of material new stock proof-clock value. An RB D0 could supply local attribution, but is not required to change the current D1 adoption/stop decision; no such attribution is claimed. Existing D0 results do not consistently dominate D1 and give no basis for the one allowed uniform D0 adjustment. No decision-changing Seed1 pair is identified, and no favorable-Seed search is opened. The optional count-to-inventory tightening is left untested here as a future source-strength question, rather than forced into a second candidate. The full potential universe remains eight configurations on six base inputs; the entered D1 pilot uses four existing inputs, not eight new instances. Stopping is based on completed full windows and independent review, not budget exhaustion.

Retain the uniform root-count D1 entry as research-only and stop further full expansion. The integer cover, nested conditional chords and scoped floor transport are valid and produce new matched LP information. Complete-method evidence is mixed: the conditional floor helps K20 relative to D0 and improves K35's incumbent, but adds no certificate over the TC2 carrier, gives no substantive proof-clock improvement over accepted stock on the two small roles, and materially regresses RB35-T2. Stable R137 GI-TBC remains the default; D0 is an attribution control, not a replacement uniform candidate. This is a finite negative adoption decision, not a claim that every count disjunction or future stronger proof must fail.

The one useful follow-up is whether a count-implied total-inventory tightening can cheaply strengthen the two conditional source floors beyond the carrier's early Native bounds. Derive and test it first on matched saved LP domains, retaining parent rows and the same physical threshold/support scope, before considering another complete panel. The current weak floors and unchanged D1-over-D0 LP optimum do not justify a branching-priority, Seed or placement grid.

6. Cost and reproducibility

New charged experimental computation is 7610.444692 s (2.114012 h), within 36000 s. Qualification and failed experimental attempts count. Complete experimental outers including postverification are also checked against the ceiling; builds, pure analysis, documents and public I/O are separately measured. Nested H/LP/MIP/Work is not charged twice. Old references keep their original clocks with zero new Optimize. Interactive reading/editing receives no invented precision.

The source/PE/DLL, inputs, configurations, exact physical/objective parameters, canonical matrices and helper identities are bound by manifests and frozen batches. Stable R137 and the R138-R140 entry/profile bytes remain unchanged. The separate count-proof switch defaults off. Local scientific validation, once-only new-carrier public byte/member readability and final Draft-PR/cleanup receipts are distinct facts. Public reading performs no Optimize, build or commercial Native load. All new assets remain under ExactEBRP-work; original user content and old rounds are protected.

References

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